

# # Next-Steps Checklist for SAM Templates

High-impact improvements to tighten security, improve resilience, and manage resources.

## ## 1) Add resilience (retries + DLQ)

Kinesis consumers: add small retries + SQS DLQ.

```
# Example
EventInvokeConfig:
  MaximumRetryAttempts: 2
  DestinationConfig:
    OnFailure:
      Type: SQS
      Destination: !GetAtt ValidationDLQ.Arn
```

## ## 2) Scope IAM to least privilege

Restrict S3 policies to prefixes, and split DynamoDB read vs. write.

```
# Example S3 prefix restriction
- Statement:
  - Effect: Allow
    Action: [ 's3:GetObject', 's3:PutObject' ]
    Resource: !Sub 'arn:aws:s3:::${Prefix}-car-bucket/records/*'
```

## ## 3) Parameterize environment & URLs

Add a Stage parameter and replace hard-coded URLs.

```
Parameters:
  Stage:
    Type: String
    Default: dev

Resources:
  JedxCarApi:
    Properties:
      StageName: !Ref Stage
      Name: !Sub '${Prefix}-car-api-${Stage}'
```

## ## 4) Control CloudWatch Logs retention

Explicit retention for Lambda log groups (e.g., 30–90 days).

## ## 5) S3 lifecycle to curb storage bloat

Expire old versions and errors after N days.

## **## 6) Structured JSON logging baseline**

Standardize logging function across Lambdas.

```
import json, logging, os
logger = logging.getLogger()
logger.setLevel(os.getenv("LOG_LEVEL", "INFO"))

def log(event, **kw):
    logger.info(json.dumps({"event": event, **kw}))
```

## **## 7) Health checks & alarms**

Add CloudWatch alarms for Kinesis lag, DLQ > 0, API 5xx errors.

## **## 8) Consistent tagging**

Add Service/Owner/Stage/CostCenter tags to all resources.